

Case of σ_x or $\sigma_y = \text{Zero}$

$$\sigma_{\max/\min} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \leq \sigma_{\text{all}}$$

$$\tau_{\max} = \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \leq \tau_{\text{all}}$$

Theories of Elastic failure :

$$\sigma_y \approx (0,5 \rightarrow 0,7) \sigma_u$$

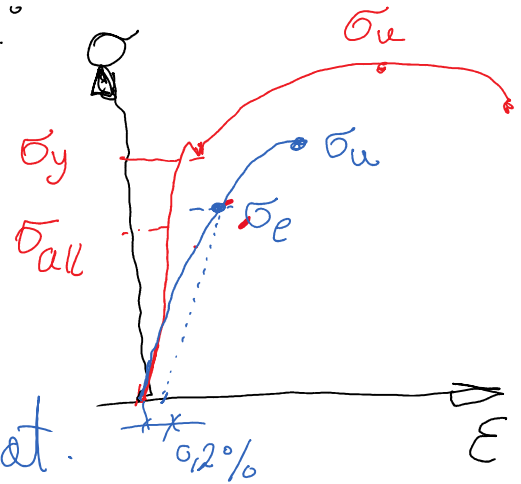
$$\sigma_e \approx (0,5 \rightarrow 0,7) \sigma_u$$

$$\boxed{\sigma_{\text{all}} = \frac{\sigma_y}{n}} \text{ for Ductile Mat.}$$

$$\boxed{\sigma_{\text{all}} = \frac{\sigma_e}{n}} \text{ for Brittle Materials.}$$

$$\tau_y = 0,5 \sigma_y \Rightarrow \boxed{\tau_{\text{all}} = \frac{\tau_y}{n} = \frac{\sigma_y}{2n}}$$

n : factor of Safety



Theories of Elastic Failure :-

1. Maximum principal Normal stress theory.
(Rankine's theory)

$$\sigma_{\max} \leq \sigma_{\text{all}}$$

2. Maximum principal Shear stress theory:
(Tresca's theory)

$$\tau_{\max} \leq \tau_{\text{all}}$$

3. Maximum principal strain theory:
(Saint Venant's theory)

$$\sigma_{\max} - \nu \sigma_{\min} \leq \sigma_{\text{all}}$$

4. Maximum principal strain Energy theory:
(Haigh's theory).

$$\sigma_{\max}^2 + \sigma_{\min}^2 - 2\nu \sigma_{\max} \sigma_{\min} \leq \sigma_{\text{all}}^2$$

5. Maximum Distortion Energy theory:
(Hencky and Von Mises theory)